Lecture 8: Bond Issuance Risk + FRA

Questions

These five multiple-choice questions follow the lecture's reasoning with different numbers. Work through the problem before reading the solutions. Keep full precision until the final answer.

USDCLP means CLP per USD. All interest rates are annual, compounded semiannually. Match forward and discount-rate maturities; ignore credit risk, fees and taxes.

1. Issue foreign-currency debt

A U.S. company sets the face equivalent of its Chilean bond at USD 2,000. USDCLP at t=0 is 1,000 CLP per USD. The bond has a 6% annual coupon and pays semiannually for two years. What are local face value and coupon?

A. CLP 2,000,000 face; CLP 60,000 per semester

B. CLP 2,000 face; CLP 60 per semester

C. CLP 2,000,000 face; CLP 120,000 per semester

D. CLP 2 face; CLP 0.06 per semester

2. Local price and dollar proceeds

For the bond in Question 1, all CLP zero rates are 8% annually with semiannual compounding. What are its local price and spot-converted USD proceeds?

A. CLP 1,927,402.10; USD 1,927,402,095.51

B. CLP 2,000,000.00; USD 2,000.00

C. CLP 2,120,000.00; USD 2,120.00

D. CLP 1,927,402.10; USD 1,927.40

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3. Build the matching forward

Today USDCLP is 1,000. The one-year CLP zero rate is 8% and the USD zero rate is 4%, both annual with semiannual compounding. What is the fair one-year forward in CLP per USD?

A. 961.908284

B. 1,040.000000

C. 1,019.607843

D. 1039.600154

4. Price the fully hedged dollar payments

Use the same bond, spot and flat CLP 8% / USD 4% curves. Convert every local payment with its matching parity forward, then discount with the USD curve. What USD price results?

A. USD 2,000.00, because face value defines price

B. USD 1,927.40, equal to local price divided by spot

C. An unknown price until future spot rates are observed

D. A higher price than local price divided by spot, because USD rates are lower

5. Interpret hedge savings

A separate CLP 1,040,000 payment is due in one year. The issuer locks the forward from Question 3. Future spot USDCLP is 800. What is the hedge saving in USD at the payment date?

A. USD 299.62 loss

B. USD 260.00 saving

C. USD 299.62 saving

D. USD 1,300.00 saving

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Worked solutions

1. Issue foreign-currency debt - Answer A

Step 1. Read USDCLP. It tells us how many pesos one dollar buys. Convert dollars into pesos by multiplying.

Step 2. Set local principal. USD 2,000 x 1,000 CLP/USD = CLP 2,000,000.

Step 3. Calculate local coupon. CLP 2,000,000 x 6% / 2 = CLP 60,000.

$$FV_{CLP} = 2000 \times 1000 = 2000000; \quad C_{CLP} = 2000000(0.06)/2 = 60000$$

Interpretation. This specifies face value. Actual issue proceeds still depend on market price.

Check your reasoning. Do not divide USD by a quote in CLP per USD.

2. Local price and dollar proceeds - Answer D

Step 1. List the peso payments. The cash flows are 60,000, 60,000, 60,000 and 2,060,000 CLP.

Step 2. Discount locally. Use 8% / 2 = 4% per semester. Their sum is CLP 1,927,402.095515.

Step 3. Convert the price today. Divide by USDCLP at t=0: 1,927,402.095515 / 1,000 = USD 1,927.402096.

$$P_{USD,0} = P_{CLP,0}/1000 = 1927.402096$$

Interpretation. The bond trades at a discount because the local yield exceeds its coupon.

Check your reasoning. Face equivalent and issue proceeds are different amounts.

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3. Build the matching forward - Answer D

Step 1. Identify base and quote currencies. USDCLP means CLP per USD: CLP is the quote currency and USD is the base currency.

Step 2. Use matching one-year growth factors. Each annual rate is divided by two and compounded for two semesters.

Step 3. Apply parity. F = 1,000 x 1.04^2 / 1.02^2 = 1039.600154 CLP per USD.

$$F_{0,1}^{USDCLP} = 1000\frac{(1+0.08/2)^2}{(1+0.04/2)^2} = 1039.600154$$

Interpretation. The higher CLP interest rate is offset by a forward exchange-rate adjustment.

Check your reasoning. The numerator uses the quote-currency rate; do not reverse the two rates.

4. Price the fully hedged dollar payments - Answer B

Step 1. Hedge each maturity. At semester k, the forward is 1,000 x (1.04/1.02)^k. Divide the peso cash flow by that quote.

Step 2. Discount the USD amount. Dividing the converted payment by 1.02^k cancels the USD growth factor embedded in the forward.

Step 3. Add the present values. Each USD PV equals its local PV / 1,000. The total is USD 1,927.40.

$$\sum_k \frac{CF_{CLP,k}/F_{0,k}}{1.02^k} = \frac{1}{1000}\sum_k \frac{CF_{CLP,k}}{1.04^k}$$

Interpretation. Different currency yields do not by themselves create free funding savings.

Check your reasoning. Do not convert every future payment using today's spot and call that a hedge.

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5. Interpret hedge savings - Answer C

Step 1. Value the unhedged payment. The stronger peso means the obligation costs 1,040,000 / 800 = USD 1,300.

Step 2. Use the locked forward for the hedge. The contracted cost is 1,040,000 / 1039.600153787 = USD 1,000.384615.

Step 3. Compare costs at the same date. Saving = 1,300 - 1,000.384615 = USD 299.62. No discounting is requested.

$$\text{Saving}_1 = \frac{1040000}{800} - \frac{1040000}{F_{0,1}} = 299.615385$$

Interpretation. A lower USDCLP makes a fixed peso payment more expensive in dollars.

Check your reasoning. Savings are unhedged cost minus hedged cost, not additional issuance proceeds.